

CHY1009
Chemistry and Environmental Studies

Module 5

Part 1

Earth - Ecosystem - Energy



Earth - Life Support System

Ecosystem Components

Energy flow in ecosystem

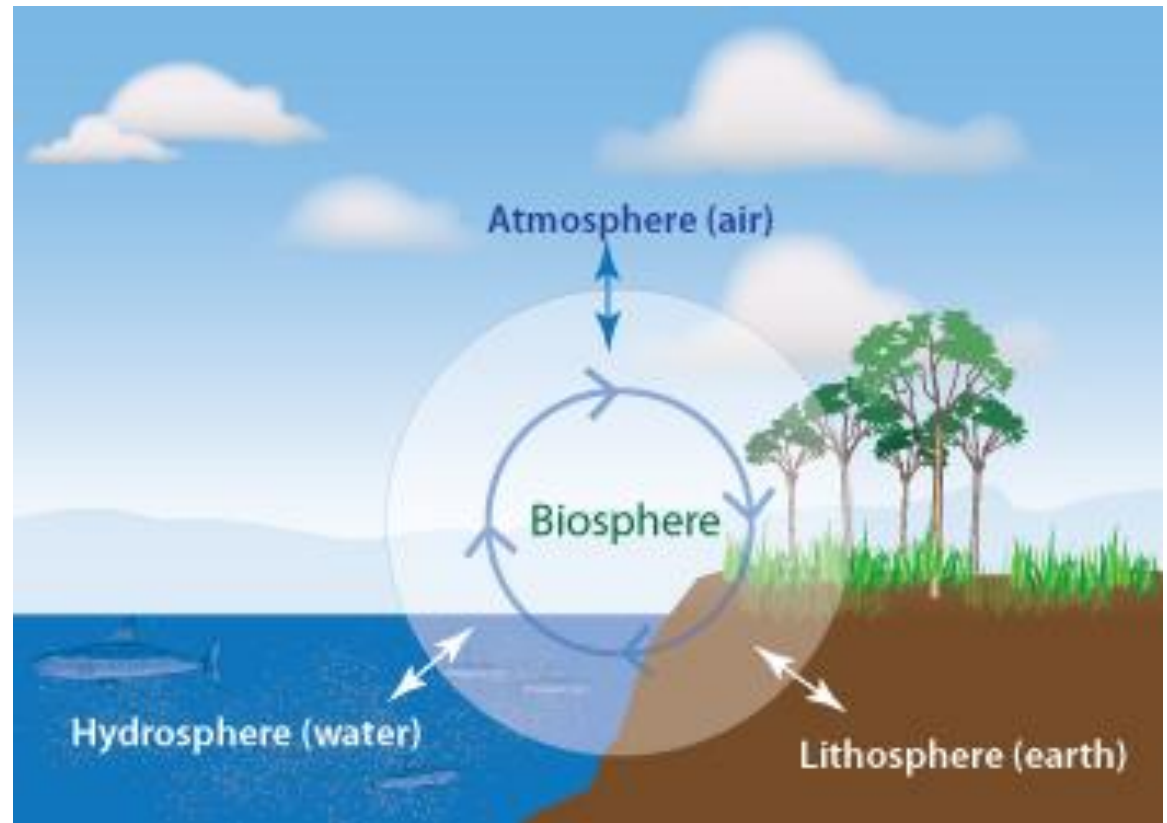
Earth's Life-Support System

The earth's life-support system consists of **four main spherical systems** that interact with one another

1. Atmosphere (air),
2. Hydrosphere (water),
3. Geosphere (rock, soil, and sediment)
4. Biosphere (living things)

Composition of Natural Environment : **Earth's Life-Support System**

- Lithosphere
- Hydrosphere
- Atmosphere
- Biosphere



One important goal of environmental science is to understand the interactions that occur within this thin layer of air, water, soil, and organisms.

Natural Environment

Lithosphere (part of geosphere!) : Earth's crust consists of the soil, rock etc. The soil is made up of inorganic and organic matters and water. Main mineral constituents are compounds derived from elements like Si, Ca, K, Al, Fe, O etc. (oxides, silicates and carbonates). Organic constituents are organo-compounds (C, H, O) of *N, P and S, and are responsible for the fertility of the soil.*

Hydrosphere : All water resources, including surface and ground water. Water is found in oceans and seas (97%), glaciers, polar icecaps (2-3%), and fresh water (lakes, rivers, streams, ground water below the land areas; **< 1%**).

Natural Environment

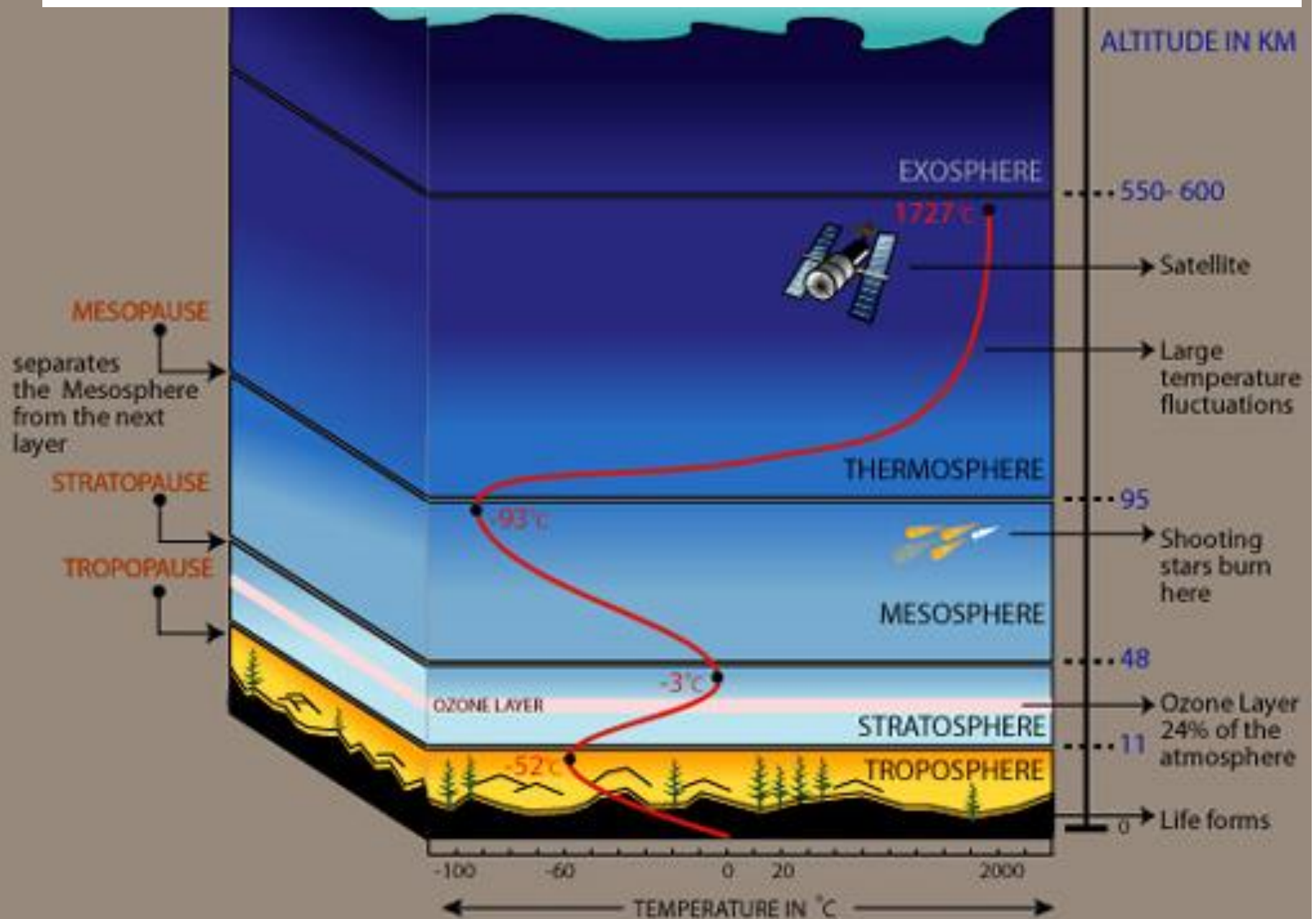
Atmosphere : *Protective* blanket of gases surrounding the earth and extends up to 500 km above the earth's surface.

Atmosphere protects earth from cosmic radiations.

Its composition include Nitrogen (78%), Oxygen (20.9%), Argon (0.9%), carbon dioxide (**0.03%**), remaining water vapour and other trace constituents.

Biosphere: Biosphere is a capsule encircling the earth's surface wherein all the *living things* exist. Life forms do not exist outside this zone. Biosphere covers parts of other segments of the environment like lithosphere, hydrosphere and atmosphere for life sustaining resources like food, water, oxygen.

Structure of Atmosphere



Earth's Life-Support System

(1) Atmosphere

The **atmosphere** is a thin spherical envelope of gases surrounding the earth's surface.

(a) Troposphere

Extends to about 17 kilometres above sea level
7 kilometers at north and south poles

It contains air that we breathe, consisting nitrogen (78%) and oxygen (21%).

Earth's Life-Support System

1. Atmosphere

(a) **Troposphere**-It contains air that we breathe, consisting nitrogen (78%) and oxygen (21%).

- The remaining 1% of the air includes water vapor, carbon dioxide, and methane, all of which are called **greenhouse gases**, which absorb and release energy that warms the lower atmosphere.
- Without these gases, the earth would be too cold for the existence of life

Earth's Life-Support System

1. Atmosphere

b. Stratosphere

17-50 kilometers above the earth's surface,

Its lower portion holds enough ozone (O₃) gas

Earth's Life-Support System

1. Atmosphere

c. Mesosphere :

The temperature decreases to -90°C at 90 km.

Coldest layer of atmosphere

Earth's Life-Support System

1. Atmosphere

d. Thermosphere:

There are not many molecules in this layer.

Air becomes less dense as we reach space.

Earth's Life-Support System

1. Atmosphere (air)

Troposphere

Stratosphere

Mesosphere

Thermosphere

2. Hydrosphere (water),

3. Geosphere (rock, soil, and sediment)

4. Biosphere (living things)

Earth's Life-Support System

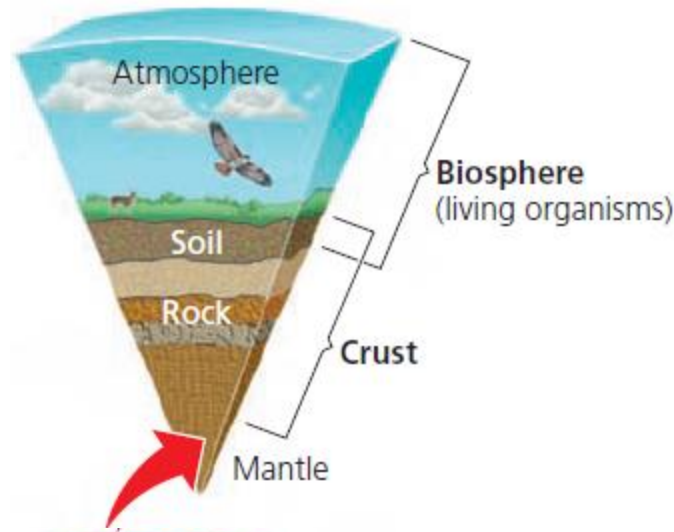
2. Hydrosphere

The **Hydrosphere** consists of all of the water on or near the earth's surface

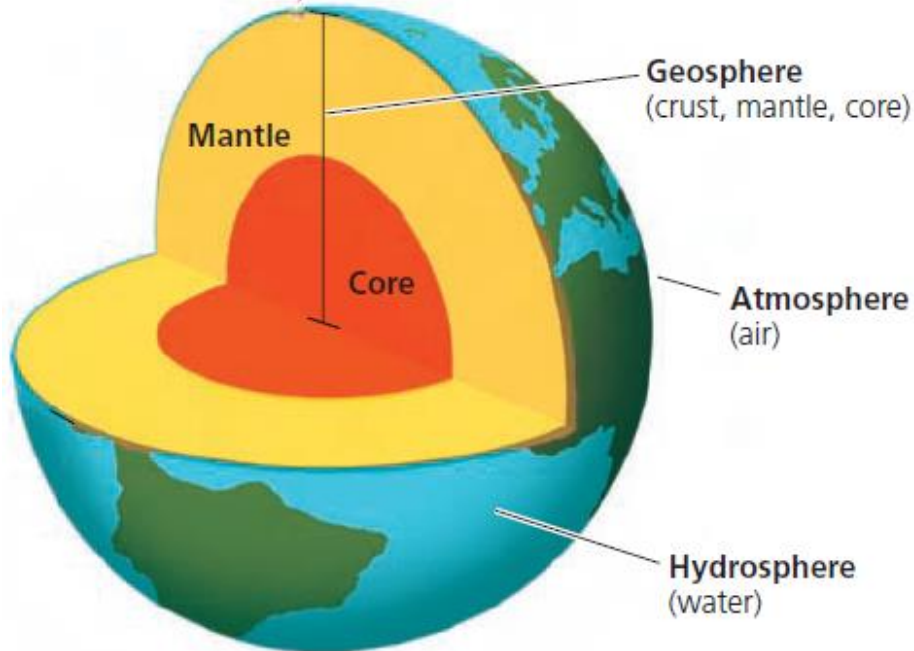
It is found as water vapor in the atmosphere, liquid water on the surface and underground, and ice—polar ice, icebergs, glaciers, and ice in frozen soil

Earth's Life-Support System

3. Geosphere

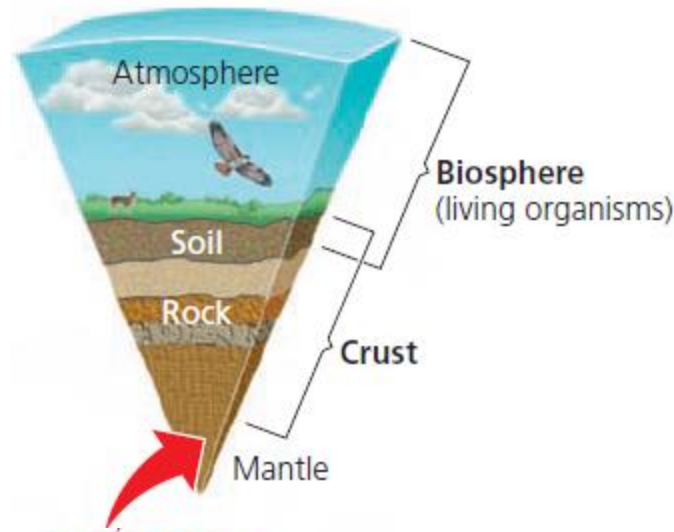


The **Geosphere** consists of the earth's intensely hot core, a thick mantle composed mostly of rock, and a thin outer crust

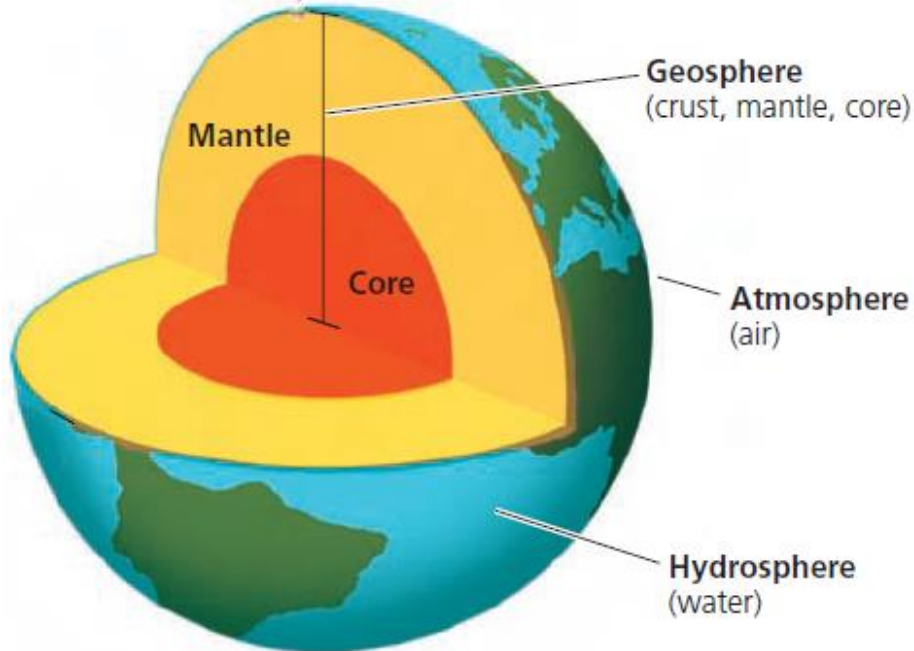


Earth's Life-Support System

4. Biosphere



The **biosphere** consists of the parts of the atmosphere, hydrosphere, and geosphere where life is found.



One important goal of **environmental science** is to understand the interactions that occur within this thin layer of air, water, soil, and organisms.

Factors Sustain the Earth's Life

1. The one-way flow of high-quality energy from the sun,
2. The cycling of nutrients
3. Biodiversity
4. Gravity

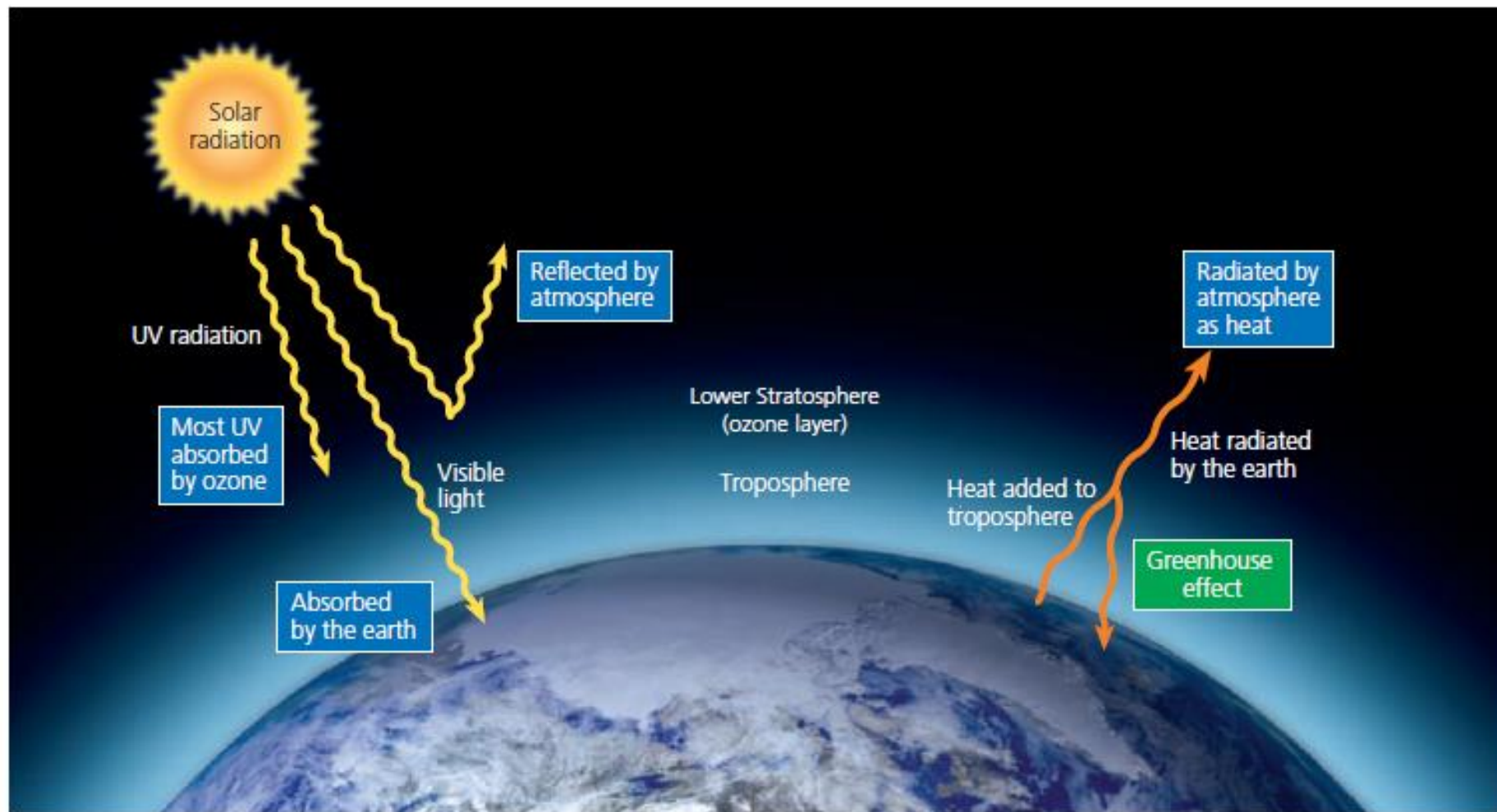
which allows the planet to hold onto its atmosphere and helps to enable the movement and cycling of chemicals through air, water, soil, and organisms.

Sun, Earth, Life, and Climate

- Only a very small amount of this output of energy reaches the earth—a tiny sphere in the vastness of space.
- The solar energy that reaches the atmosphere lights the earth during daytime, warms the air, and evaporates and cycles water through the biosphere
- Much of the high-quality solar radiation that reaches our planet is reflected by its atmosphere back into space as lower-quality energy in the form of longer-wavelength infrared radiation.

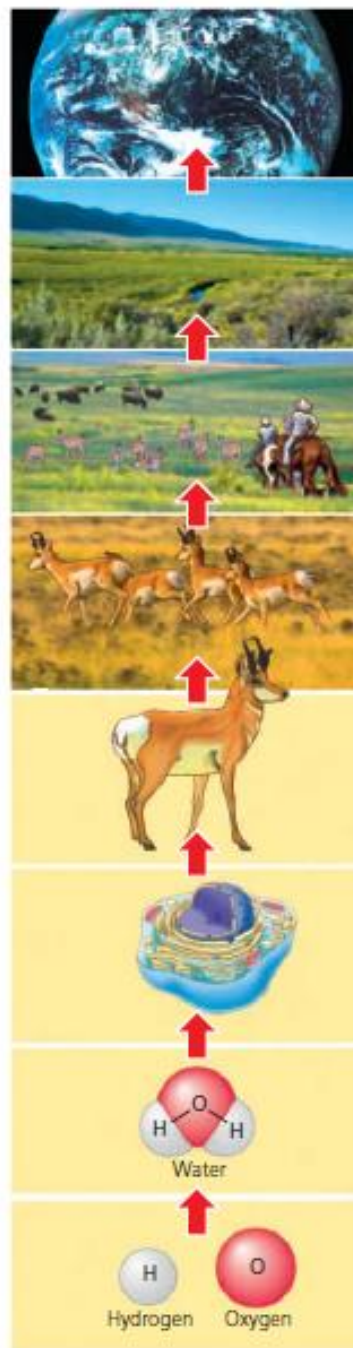
Sun, Earth, Life, and Climate

- As this infrared radiation travels back from the earth's surface into the lower atmosphere, it encounters greenhouse gases
- greenhouse gas molecules to vibrate and release infrared radiation with even longer wavelengths. The vibrating gaseous molecules then have higher kinetic energy, which helps to warm the lower atmosphere and the earth's surface.



What Are the Major Components of an Ecosystem

- Ecology is the science that focuses on how organisms interact with one another and with their non-living environment of matter and energy.
- Ecologists study interactions within and among five of these levels—
 1. organisms
 2. populations
 3. communities
 4. ecosystems
 5. biosphere



Biosphere

Parts of the earth's air, water, and soil where life is found

Ecosystem

A community of different species interacting with one another and with their nonliving environment of matter and energy

Community

Populations of different species living in a particular place, and potentially interacting with each other

Population

A group of individuals of the same species living in a particular place

Organism

An individual living being

Cell

The fundamental structural and functional unit of life

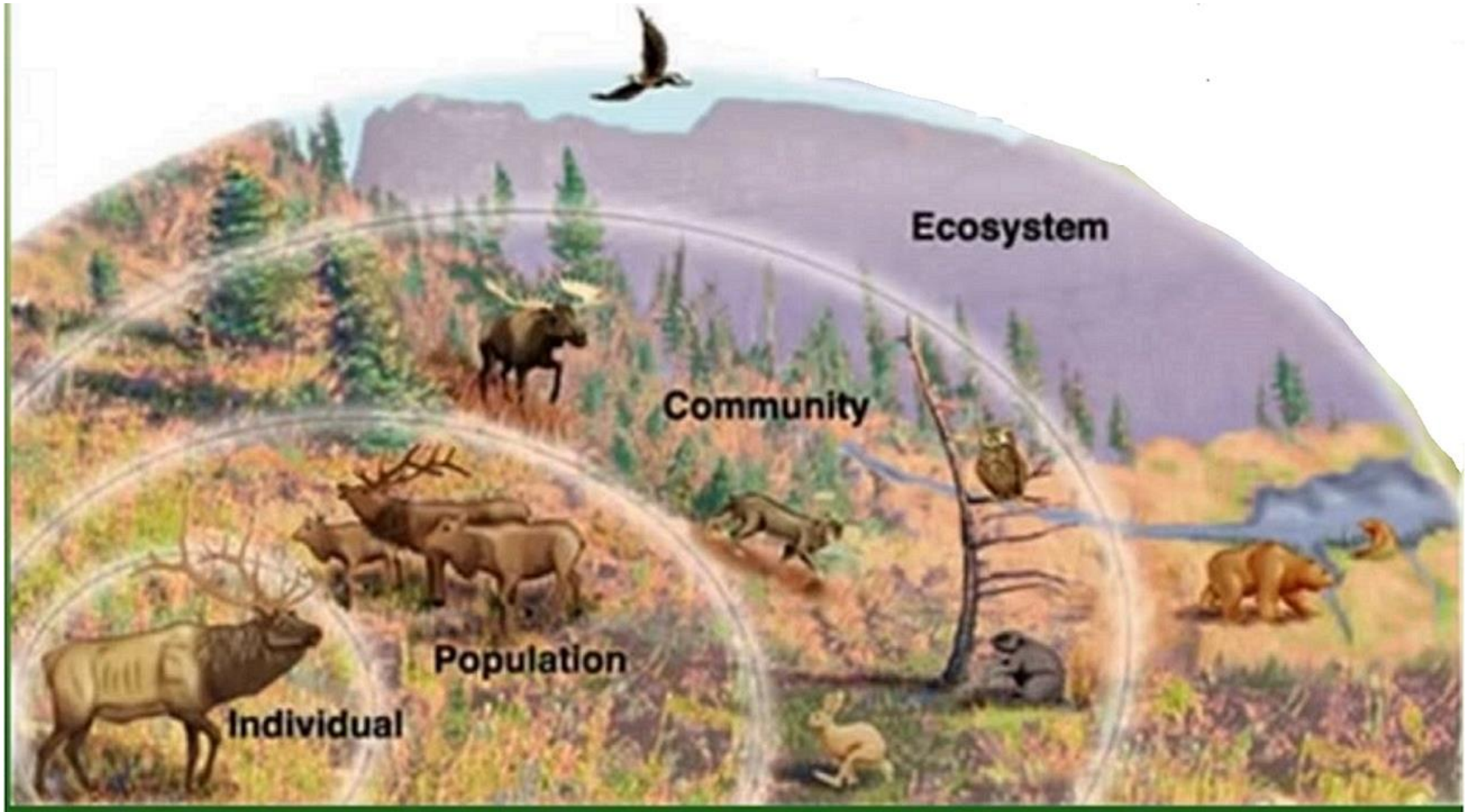
Molecule

Chemical combination of two or more atoms of the same or different elements

Atom

Smallest unit of a chemical element that exhibits its chemical properties

Levels of organizations in Ecosystem



What Are the Major Components of an Ecosystem

Ecosystems have

1. Living component (biotic)
e.g. plants, animals, microbes, and all other organisms.
2. Non-living components (abiotic)
e.g., water, air, nutrients, rocks, heat, and solar energy

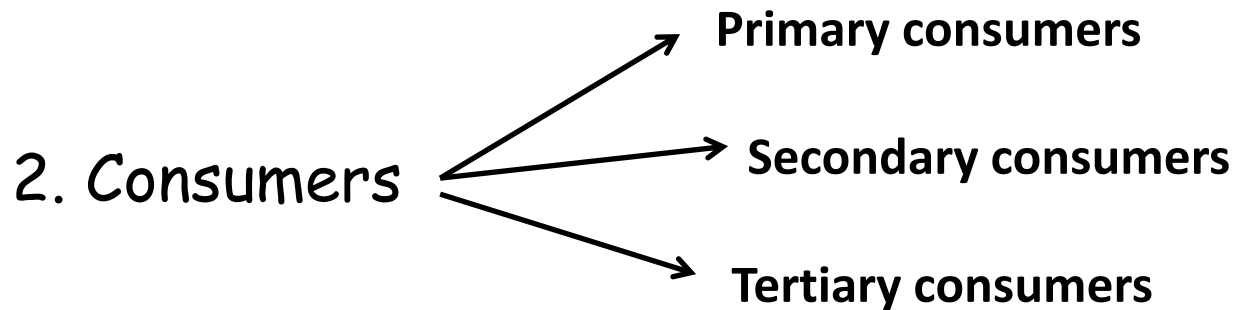
What Are the Major Components of an Ecosystem

1. Living component (biotic)

e.g. plants, animals, microbes, and all other organisms.

Ecologists assign every type of organism in an ecosystem to a **trophic (feeding) level**, depending on its source of food or nutrients.

1. Producers



3. Decomposers

What Are the Major Components of an Ecosystem

Living component (biotic)

(a) Producers

Producers, sometimes called **autotrophs** (self-feeders), make the nutrients they need from compounds and energy obtained from their environment

carbon dioxide + water + solar energy $\longrightarrow$ glucose + oxygen



Through a process called **chemosynthesis**, a few producers, mostly specialized bacteria, can convert simple inorganic compounds from their environment into more complex nutrient compounds without using sunlight.

What Are the Major Components of an Ecosystem

Living component (biotic)

(b) Consumers

- **consumers**, or heterotrophs (“other-feeders”), that cannot produce the nutrients they need through photosynthesis or other processes.
- All consumers (including humans) depend on producers for their nutrients.

Primary consumers, or herbivores (plant eaters), are animals that eat mostly green plants.

Examples are caterpillars, giraffes

What Are the Major Components of an Ecosystem

Living component (biotic)

(b) Consumers

- **Secondary consumers (or Carnivores)**, are animals that feed on the flesh of other animals

Examples are spiders, lions

- **Tertiary (or higher-level)** consumers that feed on the primary and secondary consumers.
- **Omnivores** eat both plants and other animals.

What Are the Major Components of an Ecosystem

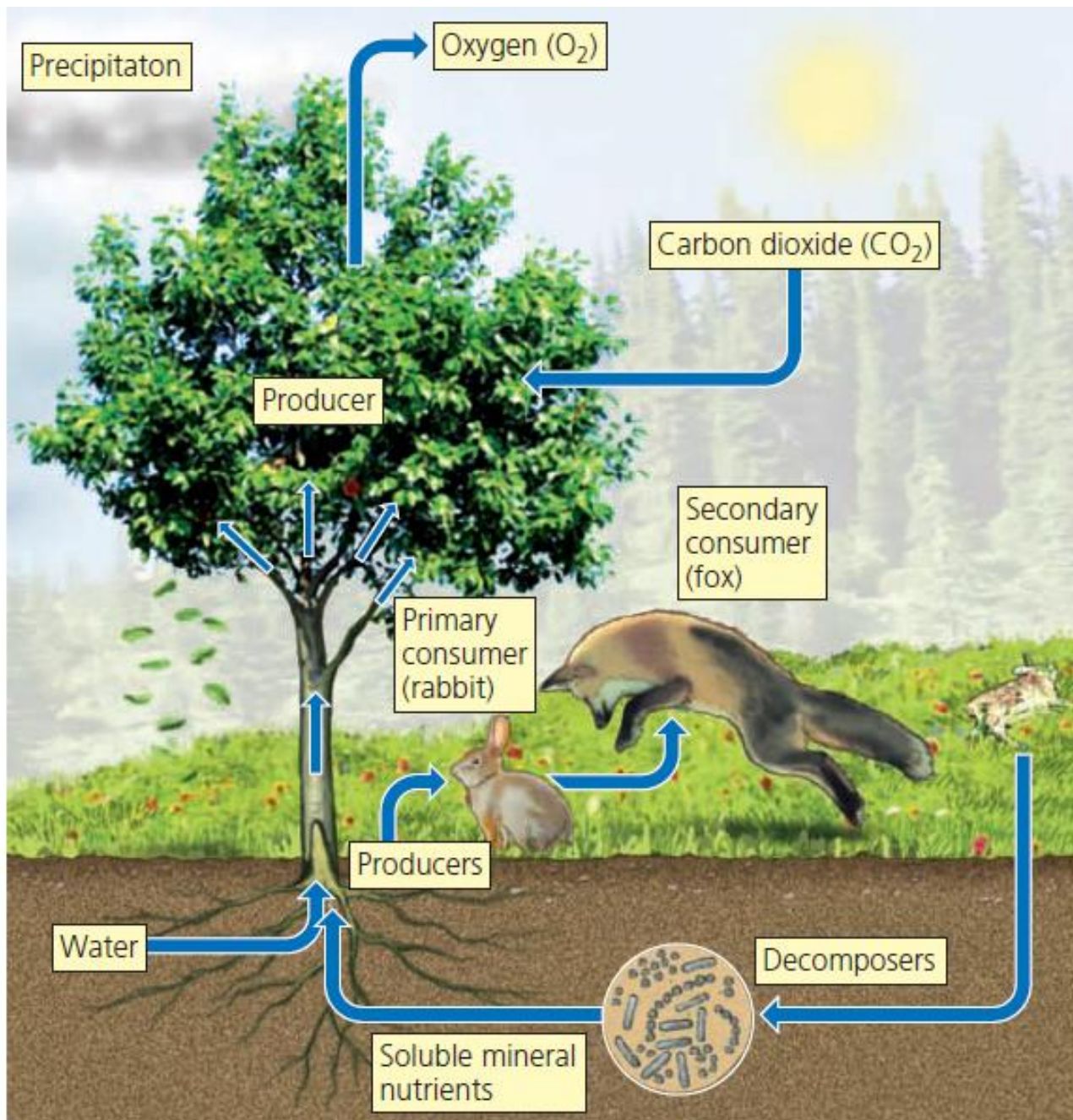
Living component (biotic)

(c) Decomposers

Decomposers (**detritivores**) are consumers that, in the process of obtaining their own nutrients, release nutrients from the wastes or remains of plants and animals and then return those nutrients to the soil, water, and air for reuse by producers

Examples are **Bacteria** and **fungi**

Without decomposers, the planet would be overwhelmed with plant litter, animal wastes, dead animal bodies, and garbage.



What Are the Major Components of an Ecosystem

Living component (biotic)

aerobic respiration

- In most cells, energy is released by **aerobic respiration**, which uses oxygen to convert glucose (or other organic nutrient molecules) back into carbon dioxide and water

glucose + oxygen $\longrightarrow$ carbon dioxide + water + energy



What Are the Major Components of an Ecosystem

Living component (biotic)

anaerobic respiration

- Some decomposers get the energy they need by breaking down glucose (or other organic compounds) in the *absence of oxygen*. This form of cellular respiration is called *anaerobic respiration*, or fermentation
- end products for this process are methane, ethanol, acetic acid and hydrogen sulfide.

Earth - Life Support System

Atmosphere (air),

Hydrosphere (water),

Geosphere (rock, soil, and sediment)

Biosphere (living things)

Ecosystem Components

Living component (biotic)

Non-living components (abiotic)

Energy flow in ecosystem

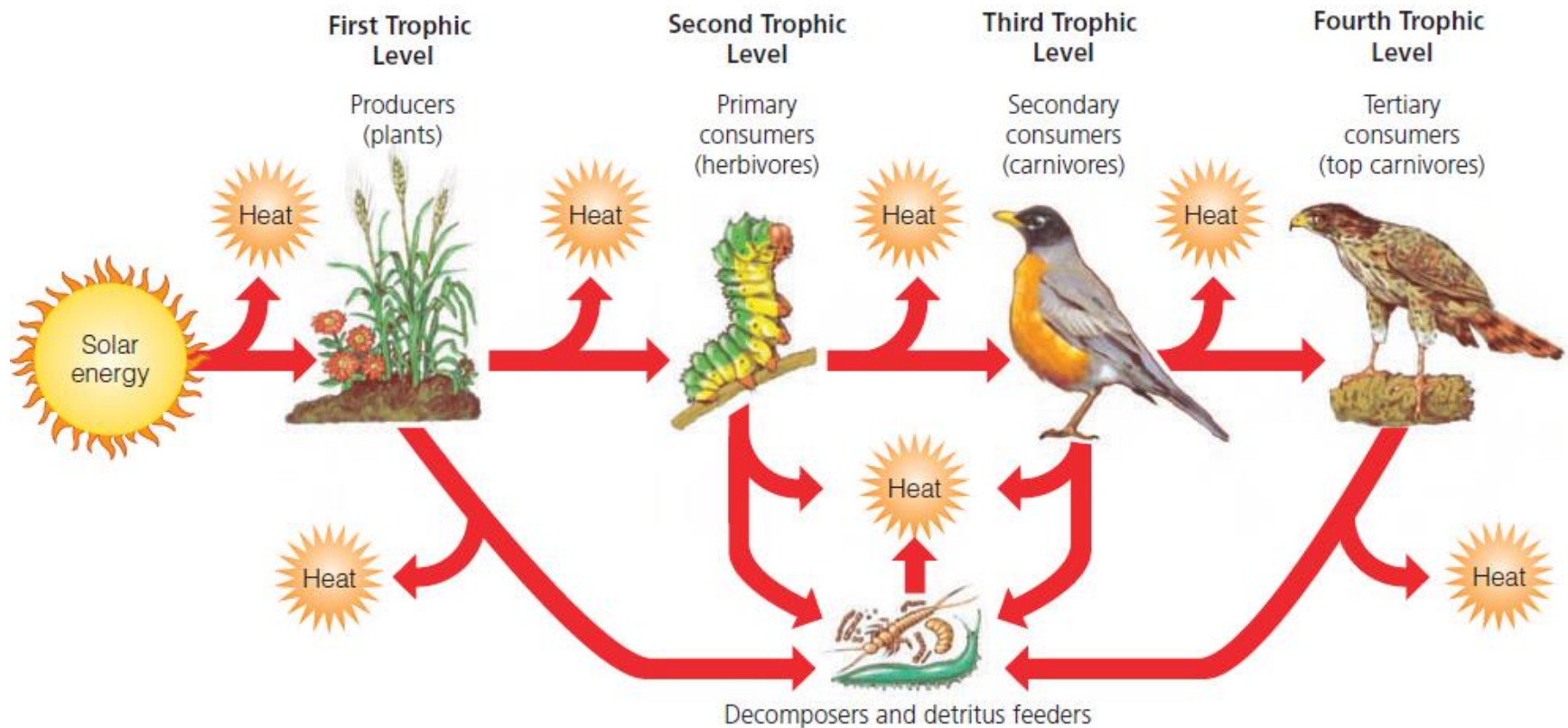
What Happens to Energy in an Ecosystem?

Energy Flows through Ecosystems in Food Chains and food webs

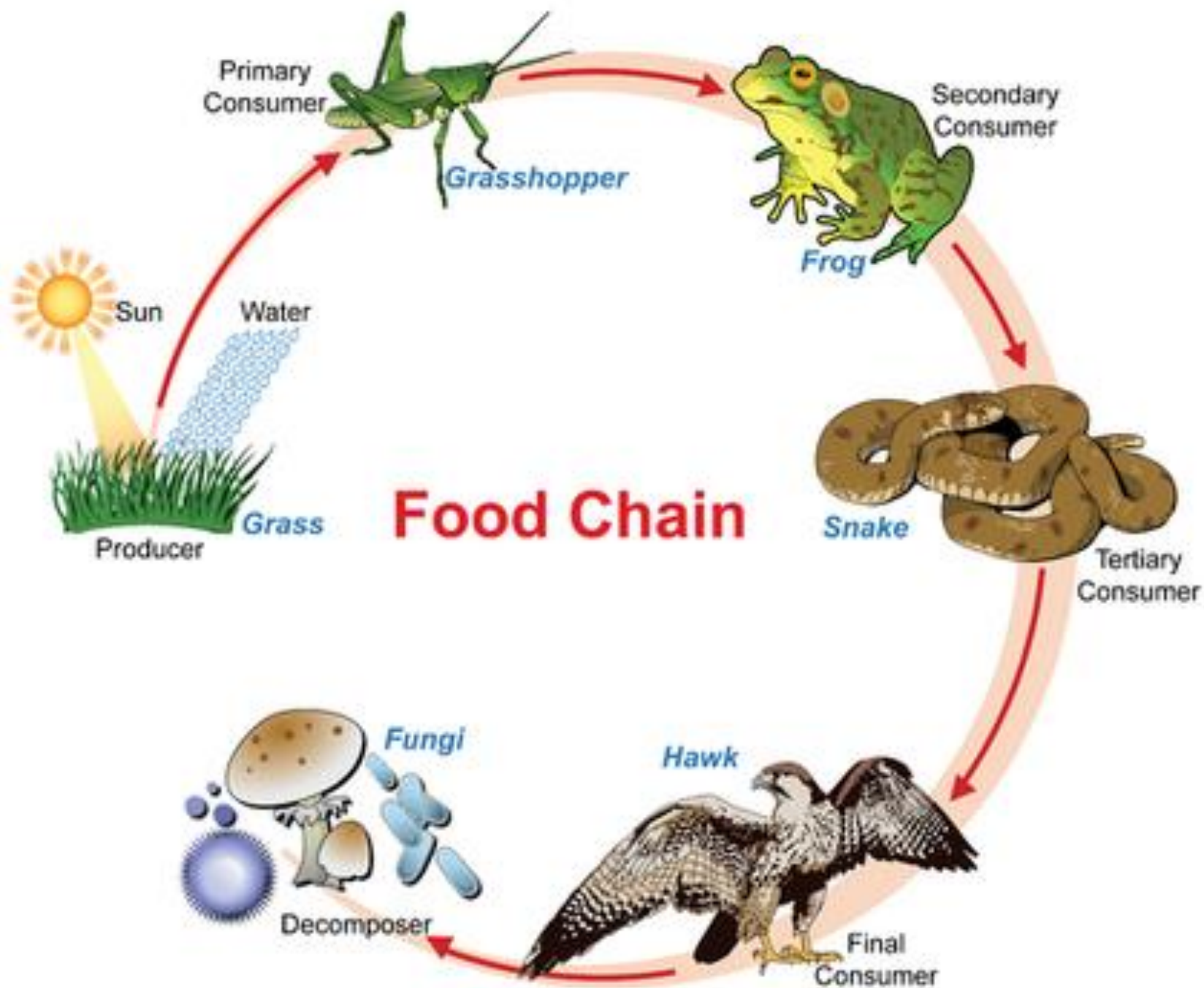
Food Chains

A sequence of organisms, each of which serves as a source of food or energy for the next, is called a **food chain**.

Food chain determines how chemical energy and nutrients move along the same pathways from one organism to another through the trophic levels in an ecosystem



Every use and transfer of energy by organisms involves a loss of some degraded high-quality energy to the environment as heat (low-quality energy)



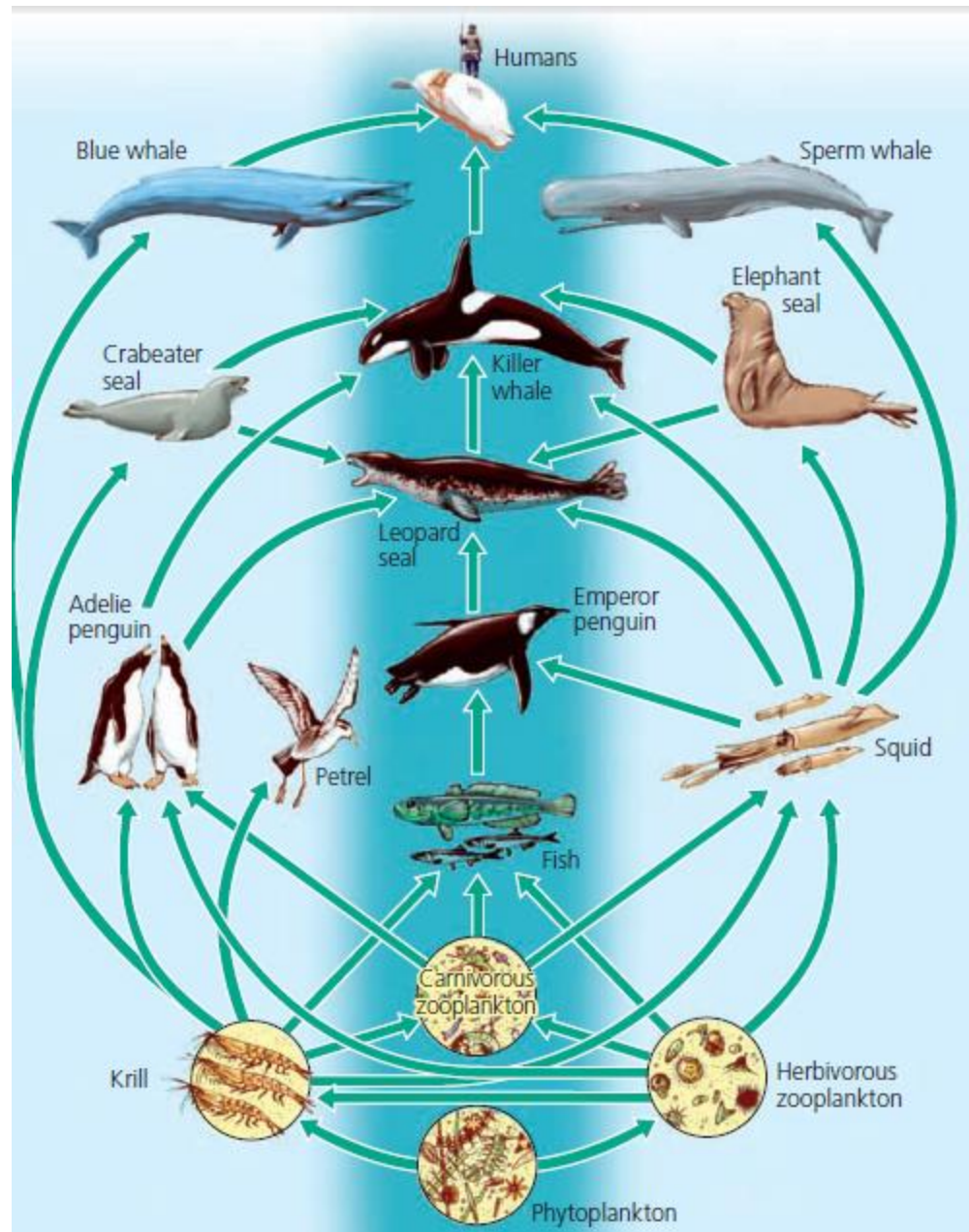
What Happens to Energy in an Ecosystem?

Energy Flows through Ecosystems in Food Chains and food webs

Food webs

organisms in most ecosystems form a complex network of interconnected food chains called a **food web**

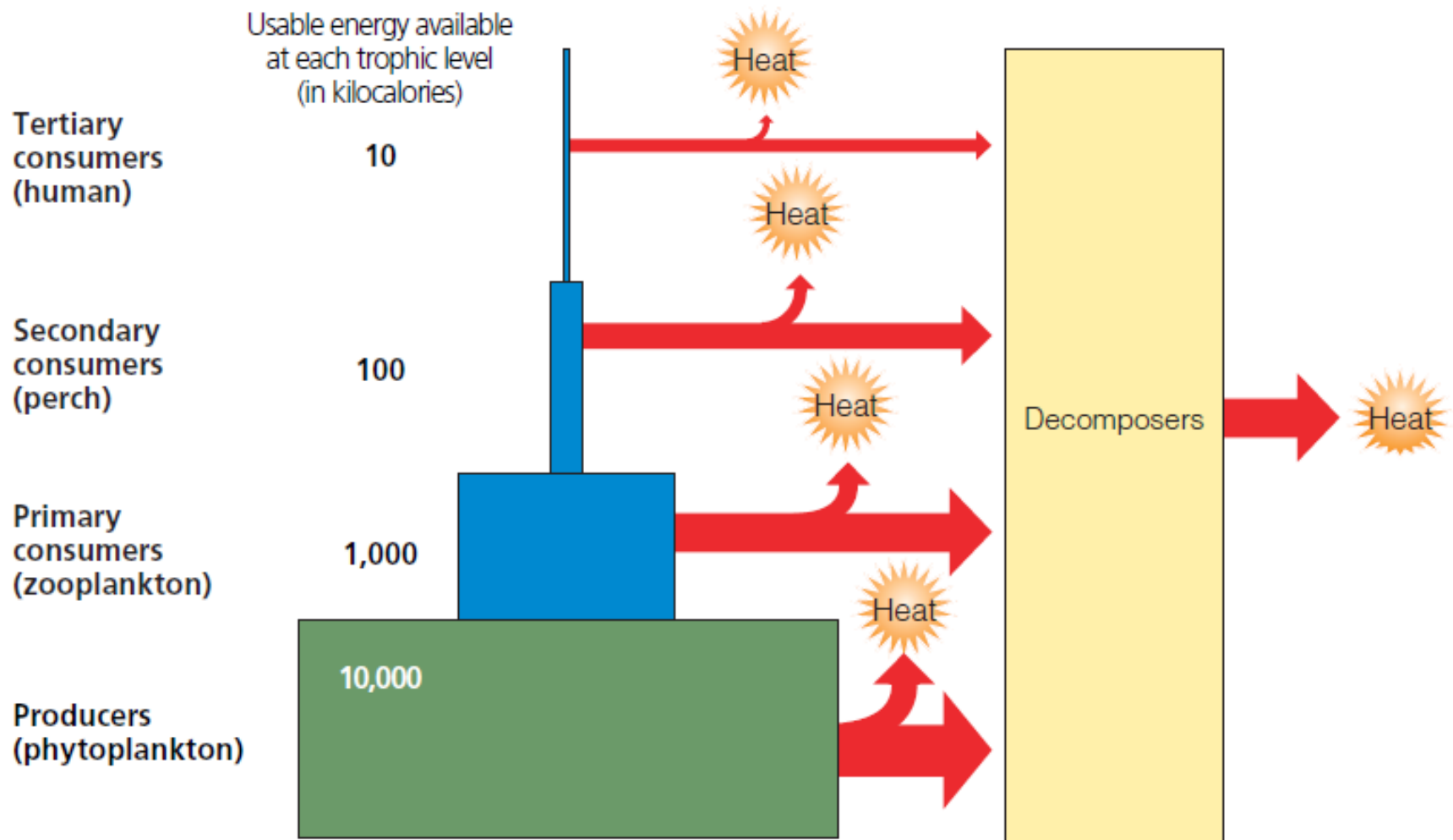
Food chains and webs show how producers, consumers, and decomposers are connected to one another as energy flows through trophic levels in an ecosystem



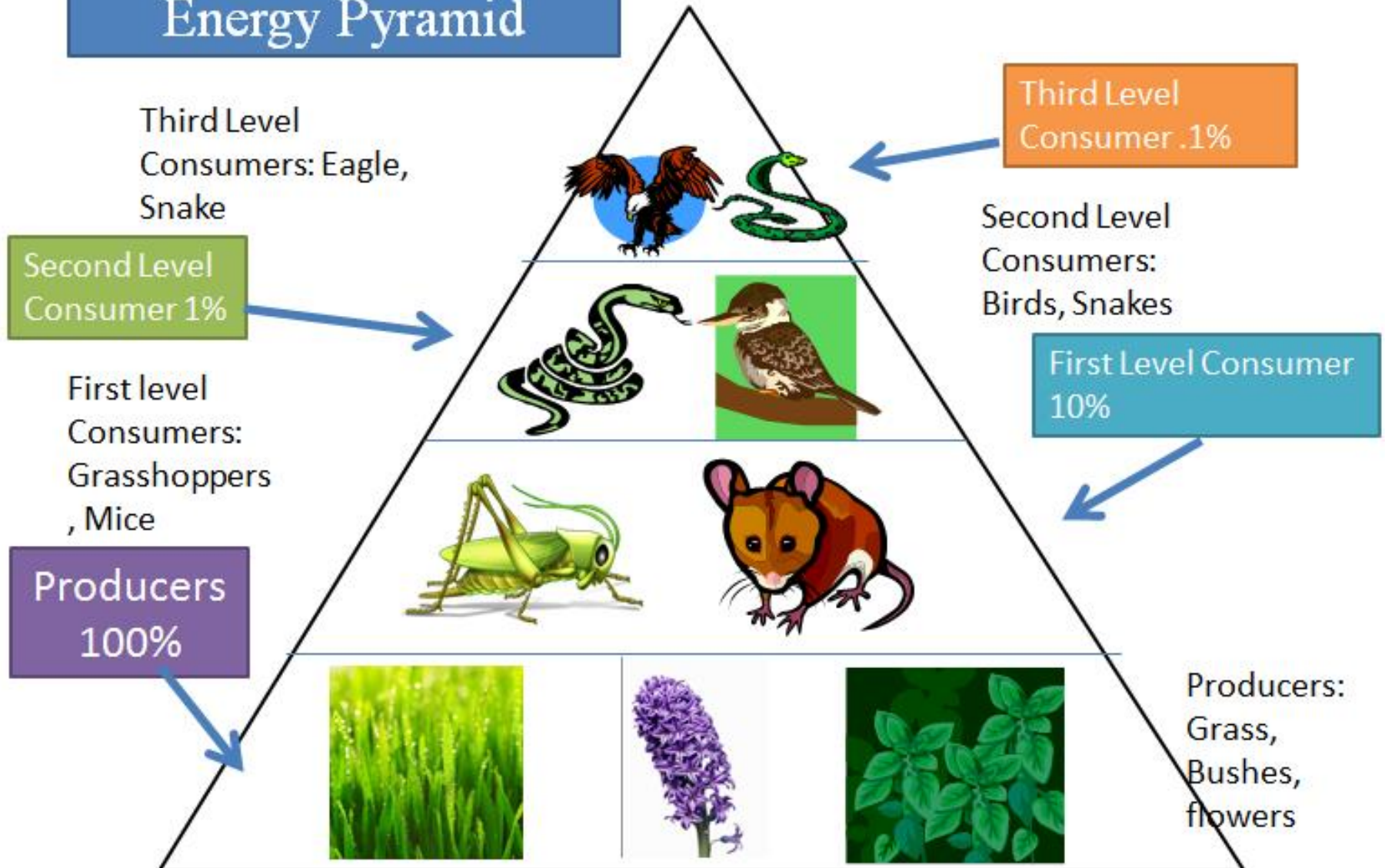
What Happens to Energy in an Ecosystem?

Usable Energy Decreases with Each Link in a Food Chain or Web

- In a food chain or web, chemical energy stored in **biomass** is transferred from one trophic level to another.
- As energy flows through ecosystems in food chains and webs, there is a **decrease in the amount of high-quality chemical energy** available to organisms at each succeeding feeding level



Energy Pyramid



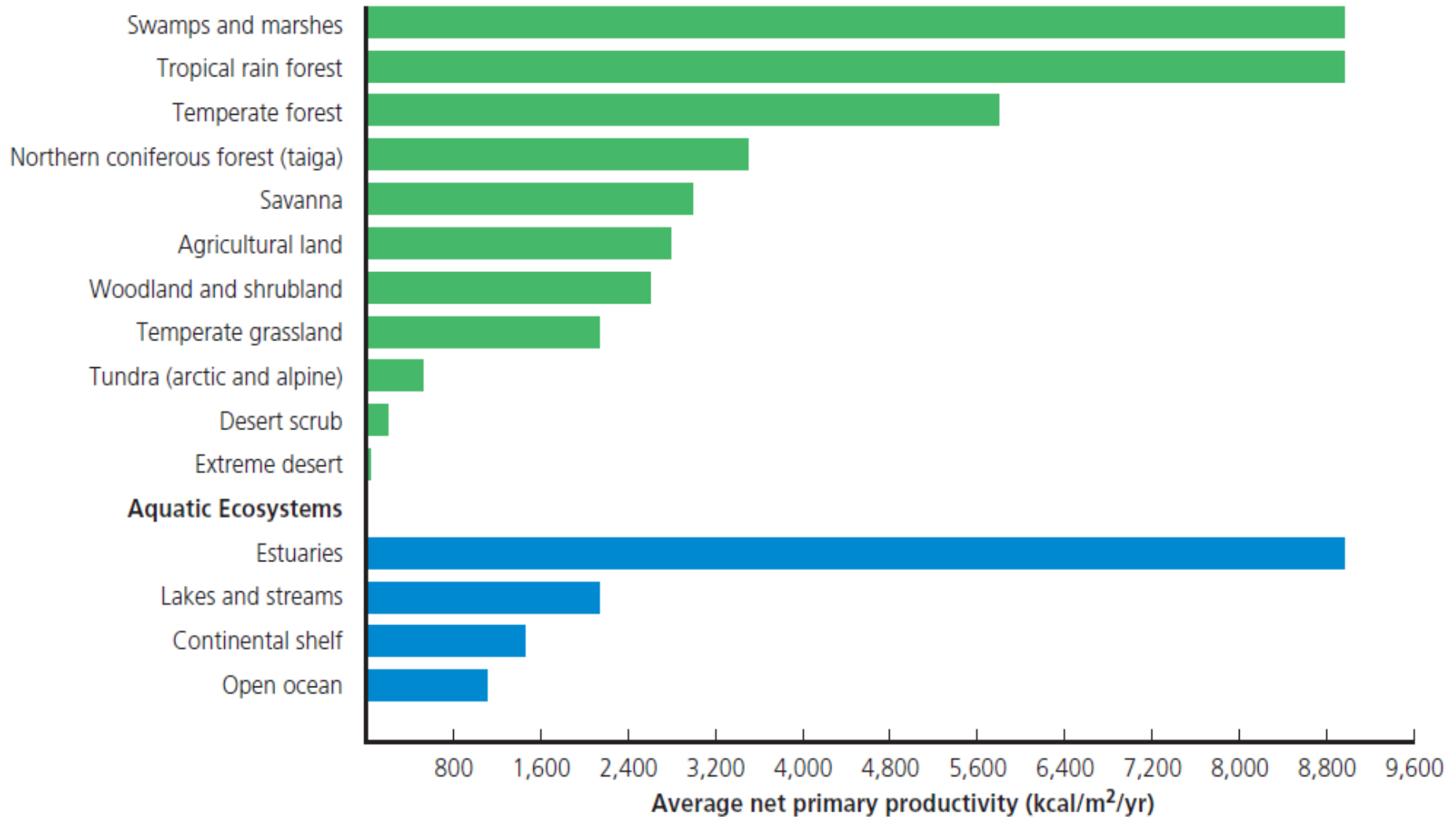
Gross Primary Productivity (GPP)

- **Gross primary productivity (GPP)** is the rate at which an ecosystem's producers (usually plants) convert solar energy into chemical energy in the form of **biomass** found in their tissues.
- GPP is usually measured in terms of energy production per unit area over a given time span, such as kilocalories per square meter per year ($\text{kcal/m}^2/\text{yr}$).

Net primary productivity (**NPP**)

- **Net primary productivity (NPP)** is the rate at which producers use photosynthesis to produce and store chemical energy minus the rate at which they use some of this stored chemical energy through aerobic respiration
- NPP measures how fast producers can make the chemical energy that is stored in their tissues and that is potentially available to other organisms (consumers) in an ecosystem.
- **NPP**, is gross **primary productivity** minus the rate of energy loss to metabolism and maintenance.
- In other words, it's the rate at which **energy is stored** as biomass by plants or other **primary** producers and made available to the consumers in the ecosystem.

Terrestrial Ecosystems



Net primary productivity (NPP)

- On land, tropical rain forests have a very high net primary productivity because of their large number and variety of producer trees and other plants.
- When such forests are cleared or burned to plant Crops, there is a sharp drop in the net primary productivity and a loss of many of the diverse array of plant and animal species.

Net primary productivity (NPP)

- As we have seen, producers are the source of all nutrients in an ecosystem that are available for the producers themselves and for the consumers and decomposers that feed on them.
- Only the biomass represented by NPP is available as nutrients for consumers, and they use only a portion of this amount.
- *Thus, the planet's NPP ultimately limits the number of consumers (including humans) that can survive on the earth.*